EO Vapour Test Cell

Selected design basis for net-transient measurement

Engineering working basis · calculation-backed concept · not released for fabrication

ACTIVE MEMBRANE	NET VOLUME / SIDE	WORKING SEGMENT
50 x 50 mm (25 cm²)	500 ± 5 mL	12 s



Figure 1. Illustrative integrated two-chamber setup. The image shows the intended architecture, not released CAD or verified internal geometry.

Selected values and reasons

Quantity	Selected value	Why it was selected
Active membrane	50 x 50 mm = 25 cm²	Largest existing aperture; strong signal and lower edge-to-area ratio.
Membrane coupon	80 x 80 mm	Provides land for double seals, guard groove and clamping.
Local gas gap	2 mm per side	Matches the local transport model; active-gap volume is only 5.0 mL per side.
Calibrated net gas volume	500 ± 5 mL per side	Retains the illustrated 0.50 L cell and slows the transient for the present instrumentation.
Bulk plenum	165 x 165 x 20 mm	Provides gross volume without changing the 25 cm² active membrane.
Polarity sequence	12 s each; +I, −I, −I, +I	Measures net transfer by polarity while suppressing fixed and linear drift.

Calculation basis

The calculation is conditional on 208 g m^{-2} h^{-1} at 32 °C, 101.325 kPa and 50% initial RH. The flux is a test-sizing target, not a measured material property.

$$\dot{m} = J\ A = 208 \times 0.0025 = 520\ \text{mg/h} = 0.144444\ \text{mg/s}$$

$$\Delta m = \dot{m}\ \Delta t = 0.144444 \times 12 = 1.733333\ \text{mg}$$

$$\Delta RH = \Delta m\ /\ (\rho sat\ Vnet) = 10.261\ \text{percentage points per chamber}$$

After the first positive segment, the ideal well-mixed endpoints are 39.739% and 60.261% RH. The differential RH signal is 20.523 percentage points.

Volume closure

$$544.5\ \text{mL gross} - 22.6\ \text{mL components} - 21.9\ \text{mL trim} = 500.0\ \text{mL net}$$

The gross 165 x 165 x 20 mm plenum is 544.5 mL. After the conservative 22.6 mL component bound, 521.9 mL remains. Removable inserts must cover 21.9–44.5 mL until actual solid displacement is measured.

Instrument and pressure implications

The SHT45 step-response fraction over 12 s is 95.0%. This does not mean exact ramp tracking: a simple first-order model indicates only 68.3% of a linear endpoint change. Use a faster calibrated reference if the transient shape is required. The rigid-cell pressure diagnostic is 976 Pa, compared with the 5 Pa target. The ideal pressure-management screen therefore requires about ±2.40 mL travel per side and 4.79 mL full stroke.

Interpretation boundary

Report: integrated net water moved through the membrane during each segment under evolving RH conditions. **Do not claim:** constant-boundary intrinsic EO flux or sustained dehumidification from the closed symmetric sequence alone. One four-segment block lasts 48 s and returns the ideal inventory to its initial state.

Fabrication remains conditional on measured mixing, leakage, fan thermal bias, differential pressure, component displacement, seal performance and assembled net volume.